

SLIDE-QUIZ EXERCISE · KEPLER01

Kepler's Laws of Planetary Motion

An Interactive Journey Through the Solar System

STUDENT STUDY GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EXERCISE OVERVIEW

Kepler's Laws of Planetary Motion

This exercise walks through the three laws of planetary motion that Johannes Kepler worked out about four hundred years ago. It opens with Kepler himself: who he was, whose observations he used, and how his laws were later explained by Isaac Newton and put to work by Edmund Halley. Each of the next three pages shows one law in action with an interactive animation and a spoken set of instructions. The First Law is about the shape of an orbit, the Second about how a planet's speed changes along it, and the Third about how a planet's distance from the Sun sets the length of its year. Every page has controls to try, and the exercise's six graded questions come straight from what the animations show.

The Exercise, Page by Page

Page	What you'll do
Introduction	A short video explains what to expect on the pages that follow.
Johannes Kepler	Read about Kepler, Tycho Brahe, Isaac Newton, and Edmund Halley. One question.
First Law	Use the eccentricity slider to change the shape of an orbit. Two questions.
Second Law	Choose a planet or Pluto and watch it sweep out equal areas; read its speed. Two questions.
Third Law	Choose up to three planets and compare their distances, periods, and T^2/a^3 . One question.
Conclusion	A closing video, your score, and the Submit Quiz button.

Learning Objectives

After this exercise you should be able to:

- ◆ Explain how Kepler used Tycho Brahe's observations, especially those of Mars, to find his laws.
- ◆ Describe how Newton's laws of motion and gravity explain Kepler's laws.
- ◆ State Kepler's First Law and describe an ellipse in terms of its foci and eccentricity.
- ◆ Define perihelion and aphelion, and identify the most and least eccentric planets.
- ◆ State Kepler's Second Law and say where in its orbit a planet moves fastest and slowest.
- ◆ State Kepler's Third Law, $T^2 \propto a^3$, and use it in years and astronomical units.
- ◆ Explain how the Transit of Venus and the Third Law together gave the distances to the planets.

QUICK REFERENCE

Key Terms

Term	Definition
Ellipse	A closed, oval curve; a “squished circle.” Every point on it has the same total distance to two fixed points, the foci.
Focus (pl. foci)	One of the two special points inside an ellipse. The Sun sits at one focus of a planet's orbit; the other focus is empty.
Eccentricity (e)	A number from 0 to just under 1 describing how flattened an ellipse is. 0 is a perfect circle; the higher the number, the flatter the orbit.
Perihelion	The point in an orbit closest to the Sun.
Aphelion	The point in an orbit farthest from the Sun.
Semimajor axis (a)	Half the longest diameter of an ellipse. For a planet it is, roughly, its average distance from the Sun.
Orbital period (T)	The time a planet takes to go once around the Sun — its year.
Astronomical unit (AU)	The average distance between Earth and the Sun, about 93 million miles.
Heliocentric model	The Sun-centered model of the Solar System proposed by Copernicus, which Kepler adopted as a student.
Transit of Venus	A rare event in which Venus passes directly across the face of the Sun as seen from Earth.
Law of Ellipses	Kepler's First Law: planets move in elliptical orbits with the Sun at one focus.
Law of Equal Areas	Kepler's Second Law: a line from the Sun to a planet sweeps out equal areas in equal times.
Law of Harmonies	Kepler's Third Law: the square of a planet's period is proportional to the cube of its semimajor axis, $T^2 \propto a^3$.
Angular momentum	A measure of rotational motion. Its conservation is the modern explanation for the Second Law.
Dwarf planet	A round body orbiting the Sun that has not cleared its orbital neighborhood. Pluto is one.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
1571-1630	Kepler's lifetime
1609-1619	Years in which Kepler published his three laws
~93 million mi	One astronomical unit (AU), the average Sun-Earth distance
6	Planets known in Halley's time, Mercury through Saturn
0.00 - 0.80	Range of the eccentricity slider on the First Law page
0.007	Venus's eccentricity — the lowest of the planets
0.017	Earth's eccentricity — nearly a circle
0.093	Mars's eccentricity, which let Kepler see the orbit was not a circle
0.206	Mercury's eccentricity — the highest of the planets
0.248	Pluto's eccentricity, higher than any planet's
$T^2 / a^3 \approx 1$	Third Law ratio for any planet, with T in years and a in AU
1 yr, 1 AU	Earth's period and semimajor axis, the units the Third Law uses

EXERCISE CONTENT

Page-by-Page Summary

Part 1 — Johannes Kepler

Johannes **Kepler** (1571–1630) was a German mathematician, astronomer, and astrologer. As a student at the University of Tübingen he became a devoted follower of the Sun-centered, or **heliocentric**, model of Copernicus, and despite poverty, religious persecution, and the loss of family members he kept at his astronomical work all his life.

Kepler derived his three laws by hand from the careful observations of his colleague **Tycho Brahe**. For the First Law, that planetary orbits are **ellipses** rather than circles, he relied mainly on Brahe's data for Mars. The laws were published between 1609 and 1619.

Kepler's laws describe how the planets move; they do not say why. Decades later, **Isaac Newton** derived all three from his more fundamental Three Laws of Motion and his Law of Gravity. Because Kepler's laws matched the observations, they served as the experimental confirmation of Newton's theory.

Newton's friend **Edmund Halley** saw how to turn the Third Law into real distances. The Third Law gives every planet's distance in **astronomical units**, the average Sun–Earth distance. Measure the AU itself — using the rare **Transit of Venus** and trigonometry — and the distances to all six known planets follow. The AU is now known to be about 93 million miles.

Kepler also contributed to optics, discovered two new regular polyhedra, and provided a theoretical basis for logarithms. His work is a bridge between ancient and modern astronomy, combining careful mathematics with observation.

Part 2 — The First Law: The Law of Ellipses

Planets move in elliptical orbits with the Sun at one focus. An ellipse has two **foci**; the Sun occupies one, and the other is empty. So a planet's distance from the Sun changes as it goes around. Its closest point is **perihelion** and its farthest is **aphelion**.

How flattened an ellipse is is measured by its **eccentricity**. In the animation the slider runs from 0.00 to 0.80. At 0.00 the two foci merge at the center and the orbit is a perfect circle. As the eccentricity rises, the orbit keeps the same length from left to right but gets narrower from top to bottom, and the Sun's focus slides toward one end.

At high eccentricity the planet's speed is obviously not constant: it races around the Sun's end of the orbit and creeps along the far end. That observation leads directly to the Second Law.

Real planetary orbits are close to circles. Venus (0.007) and Neptune (0.009) are almost perfectly round, Earth is 0.017, and Mercury, at 0.206, is the most eccentric planet.

Part 3 — The Second Law: The Law of Equal Areas

A line connecting a planet to the Sun sweeps out equal areas in equal time intervals. Near the Sun that line is short, so the planet must move quickly to cover a given area; far from the Sun the line is long, so a slower motion covers the same area. Planets therefore move fastest

at **perihelion** and slowest at **aphelion**. Today this is understood as a consequence of the conservation of **angular momentum**.

In the animation you can select any of the eight planets or the **dwarf planet** Pluto. Its eccentricity appears in the upper left, along with its current speed in miles per second and kilometers per second. The lowest speed flashes in red and the highest in green.

The more eccentric the orbit, the bigger the difference between those two speeds. For Pluto (eccentricity 0.248) the area chunks swept out are visibly different shapes — short and wide near the Sun, long and thin far away — yet their areas are equal.

Part 4 — The Third Law: The Law of Harmonies

The square of a planet's orbital period is proportional to the cube of its semimajor axis: $T^2 \propto a^3$. With the period T in Earth years and the semimajor axis a in astronomical units, $T^2/a^3 \approx 1$ for every planet. Planets farther from the Sun take much longer to go around.

The animation lets you choose up to three planets at a time; choosing a fourth removes the first one chosen (first in, first out). For each planet it shows a , T , and the ratio T^2/a^3 , so you can check that the ratio stays close to 1. For simplicity, the orbits are drawn as circles.

Kepler's Third Law does not say so directly, but it follows that the farther a planet is from the Sun, the slower it moves. Mercury, closest to the Sun, is the fastest planet.

This is the law Halley used. Once the AU was measured with the Transit of Venus, each planet's a , already known in AU from the Third Law, could be converted into miles.

TRY IT IN THE ANIMATIONS

On the First Law page, move the slider slowly from 0.00 to 0.80 and watch which dimension of the orbit changes. On the Second Law page, compare the red and green speeds for Venus, Earth, Mercury, and Pluto. On the Third Law page, pick Mercury, Earth, and Saturn together and check T^2/a^3 for each.

STUDY QUESTIONS

Questions for Review

Answer in your own words. Where a question asks you to explain rather than to state, a one-sentence answer is unlikely to be complete.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — Johannes Kepler

1. Whose observations did Kepler use to find his laws, and which planet's data mattered most for the First Law?

2. Which model of the Solar System did Kepler follow, and where did he come to accept it?

3. How are Kepler's laws connected to Isaac Newton's work?

4. What is an astronomical unit, and what rare event was used to measure it?

5. Besides planetary motion, name two of Kepler's other contributions.

Part 2 — The First Law: The Law of Ellipses

1. State Kepler's First Law.

2. An ellipse has two foci. What is at each focus of a planet's orbit?

3. Define perihelion and aphelion.

4. What does eccentricity measure? Describe the orbit in the animation at 0.00 and at 0.80.

5. Using the eccentricity table, which planet has the most eccentric orbit and which the least?

6. With the slider at its maximum, how does the planet's speed change as it goes around?

Part 3 — The Second Law: The Law of Equal Areas

1. State Kepler's Second Law.

2. At what point in its orbit does a planet move fastest, and where does it move slowest?

3. Use the idea of equal areas to explain why a planet must move faster when it is close to the Sun.

4. For Pluto, do the area chunks swept out have the same shape? Do they have the same area?

5. Why does a more eccentric orbit show a bigger difference between the highest and lowest speeds?

6. What modern physical principle explains the Second Law?

Part 4 — The Third Law: The Law of Harmonies

1. State Kepler's Third Law in words and as a proportion.

2. In what units does T^2/a^3 come out close to 1, and what do T and a stand for?

3. Jupiter's semimajor axis is about 5.2 AU. Use $T^2 = a^3$ to estimate its orbital period.

4. Which planet moves fastest around the Sun, and how does a planet's speed depend on its distance?

5. Why was measuring the AU the missing piece Halley needed to find the distances to the planets?

6. The Third Law animation draws every orbit as a circle. Why is that a reasonable simplification?

Notes